Jiajun Hu

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Summary

Ph.D. student in Computer Engineering working on FPGA architecture & CAD, in-memory computing, and DRAM-PIM for ML/LLM inference. Skilled in Verilog/SystemVerilog, Synopsys DC, VTR/VPR (C++), and Python performance, energy, and area modeling.

Education

Arizona State University — Tempe, AZ
Ph.D. in Computer Engineering — 08/2024 – Present

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Electrical and Computer Engineering — 02/2022 – 12/2024

University of Nottingham, Ningbo, China — Ningbo, China
Bachelor of Engineering in Electrical and Electronics Engineering — 09/2017 – 07/2021

Industry Experience

Research Associate Intern — HPE Labs — Fort Collins, CO
Emerging Accelerator Team — 05/2026 – Present

- Developing FPGA–ReRAM in-memory-computing chiplet architectures for LLM inference.
- Building system-level performance, energy, and area models for chiplet design-space exploration.

Research Experience

Graduate Research Assistant — Arizona State University — Tempe, AZ
FPGA architecture and CAD, in-memory computing — 08/2024 – Present

NIFA: Nonlinear IMC Enhanced FPGA for Efficient ML Inference

International Conference on Field-Programmable Logic and Applications (FPL), 2026

- Proposed ADC-free ReRAM IMC blocks for FPGAs, with analog CAMs (ACAMs) for nonlinear ops and log-domain attention.
- Evaluated the architecture in VTR on ResNet-9, VGG-11, and BERT-Tiny benchmarks (Fmax, wirelength, area).
- Ran FPGA-aware DSE of crossbar size and IMC area budget.
- Extended a Python IMC-FPGA simulator for end-to-end performance, energy, and area.
- Achieved up to 40×/1.9× energy and 4.1×/2.5× area efficiency gains over the prior SOTA on CNN and Transformer workloads.
- Built a Verilog IP library (GEMM, dynamic MatMul, online softmax, attention), cycle-matched to Python and NumPy models.

Direct BRAM-DSP Paths for Deep Learning on FPGAs

International Conference on Field-Programmable Technology (FPT), 2026

- Proposed hardened BRAM-to-DSP direct paths that bypass global routing, at negligible overhead (Synopsys DC, FreePDK45).
- Extended the VPR placer (C++) with cross-type placement macros and connectivity-weighted placement priority.
- Built a Python place-and-route evaluation flow and evaluated the architecture on FC, Conv, attention, and LSTM benchmarks.
- Achieved up to 25% higher Fmax and 49% lower wirelength on DL designs.

RACAM: Reuse-Aware Bit-Serial DRAM-PIM with Automated Mapping

International Conference on Parallel Architectures and Compilation Techniques (PACT), 2026

- Built the initial bit-serial DRAM-PIM GEMM/GEMV performance model on LLMCompass (Python).
- Designed the initial tiling and mapping search (array/bank/device tiling, broadcast, loop order).
- Implemented bit-serial integer PEs and bank/rank/device broadcast/mux logic in SystemVerilog; synthesized with Synopsys DC.
- Built the evaluation pipeline for GPT-3 and Llama-3.1 prefill/decode against GPU baselines.

Publications

- **Jiajun Hu**, Ruthwik Reddy Sunketa, Lei Zhao, Archit Gajjar, Luca Buonanno, and Aman Arora. *NIFA: Nonlinear IMC Enhanced FPGA for Efficient ML Inference*. In *International Conference on Field-Programmable Logic and Applications (FPL)*, 2026. arXiv:2607.15123
- **Jiajun Hu**, Ruthwik Reddy Sunketa, Andrew Boutros, and Aman Arora. *Boosting FPGA Performance with Direct BRAM-DSP Paths*. In *International Conference on Field-Programmable Technology (FPT)*, 2026 (short paper). arXiv:2607.05756
- Siyuan Ma, **Jiajun Hu**, Jeeho Ryoo, Aman Arora, and Lizy Kurian John. *RACAM: Enhancing DRAM with Reuse-Aware Computation and Automated Mapping for ML Inference*. In *International Conference on Parallel Architectures and Compilation Techniques (PACT)*, 2026. arXiv:2512.09304
- **Jiajun Hu**, Chetan Choppali Sudarshan, Maxwell Clifford, Vidya Chhabria, and Aman Arora. *CarbonSet: A Dataset to Analyze Trends and Benchmark the Sustainability of CPUs and GPUs*. In *Great Lakes Symposium on VLSI (GLSVLSI)*, 2025, pp. 177–184. doi:10.1145/3716368.3735235
- Chetan Choppali Sudarshan, **Jiajun Hu**, Aman Arora, and Vidya A. Chhabria. *CarbonPATH: Carbon-aware pathfinding and architecture optimization for chiplet-based AI systems*. arXiv preprint (under review). arXiv:2603.03878